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How and when do virtual influencers positively affect consumer responses to endorsed brands?

Barbara Kobuszewki Volles [a,*], Jaewoo Park [b], Anneleen Van Kerckhove [a], Maggie Geuens [a]

[a] Ghent University, Department of Marketing, Innovation and Organisation, Tweekerkenstraat 2, 9000 Gent, Belgium
[b] Chuo University, Department of Marketing and International Trade, Faculty of Commerce, 742-1 Higashinakano Hachioji-shi, Tokyo, Japan

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ABSTRACT

While companies' interest in virtual influencers as brand endorsers is growing, it is striking that limited research exists on how virtual influencers shape consumer perceptions and responses to advertising and the brands they endorse. Scholars have predominantly highlighted the drawbacks of utilizing virtual influencers. Differing from this literature, we explore virtual influencers' advantages or positive attributes which may lead to successful virtual influencer endorsements. Furthermore, while virtual influencers may be seen as less trustworthy than humans, we posit that their perceived uniqueness represents a pivotal advantage. Four studies show that greater perceived uniqueness holds the potential to mitigate trustworthiness concerns and lead to positive outcomes, especially when consumers strongly identify with virtual influencers.

1. Introduction

In recent years, virtual influencers have become immensely popular on social media. These digital personalities have demonstrated their ability to generate higher audience engagement than human influencers, boasting an average engagement rate[1] of 2.84 % compared to 1.72 % for human influencers (Geyser, 2024). This underscores their effectiveness in capturing the audience's attention, and potentially generating interest in the products and brands they endorse. Virtual endorsers have even become brand ambassadors, securing partnerships with renowned companies. For instance, @imma.gram, with 399 k followers, has a remarkable 2.21 % engagement rate and has already partnered with well-known brands like Hugo Boss, Rakuten, Ikea, and Prada. However, despite the growing interest in virtual influencers as a unique type of brand endorser, our understanding of how they influence consumer responses to advertising and the advertised brand remains limited.

Research has predominantly focused on the drawbacks associated with employing virtual influencers. For instance, virtual influencers are often considered less effective endorsers than their human counterparts. This is primarily due to perceived limitations in sensory capabilities and credibility, except if the sensory cue salience in the ad is diminished (Li et al., 2023). Additionally, consumers generally hold less positive attitudes toward virtual influencers than human influencers largely because the former are, amongst others, perceived as less warm and trustworthy (Wan & Jiang, 2023). Compared with human influencers, virtual influencers are frequently seen as less trusted sources because of a perceived greater social distance (Sands et al., 2022). Differing from extant research, which predominantly highlights the disadvantages of virtual (versus human) influencers, we explore their potential advantages and the circumstances under which they can positively impact consumer responses to endorsed brands.

Specifically, we aim to understand how and when virtual influencers positively affect consumer responses to endorsed brands. We conduct four studies to investigate the factor which may shed light on virtual influencers' effectiveness in advertising campaigns. While we acknowledge that virtual (versus human) influencers may be perceived as less trustworthy, we argue that their perceived uniqueness can potentially mitigate trustworthiness issues' negative impact and even lead to positive outcomes, especially when consumers have a strong sense of identification with these virtual influencers.

Our research contributes to the literature on virtual influencers in two significant ways. First, contrary to prior research focused on

* Corresponding author.
E-mail addresses: barbara.kobuszewskivolles@ugent.be (B.K. Volles), park001@g.chuo-u.ac.jp (J. Park), anneleen.vankerckhove@ugent.be (A. Van Kerckhove), maggie.geuens@ugent.be (M. Geuens).

[1] The engagement rate measures how actively an influencer's social media audience engages with their content. It is calculated as a percentage by dividing the interactions (likes, comments, shares, etc.) by the influencer's total followers and then multiplying by 100. This metric indicates an influencer's capacity to capture attention for the promoted products or services in their posts, signifying their potential to deliver results for a brand (Alain, 2024).

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negative perceptions towards virtual (versus human) influencers (e.g., Franke et al., 2022; Sands et al., 2022), we introduce perceived uniqueness as a positive perception. Specifically, it leads to more favorable behavioral outcomes towards virtual (versus human) influencers. Additionally, we explore and demonstrate ambivalent perceptions toward virtual influencers in consumer minds. These findings contribute to a deeper and more nuanced understanding of consumer responses to virtual influencers. Second, we highlight consumer-endorser identification's significance in influencer marketing, emphasizing its role in driving positive consumer responses to both human (e.g., Carlson et al., 2020; Schouten et al., 2019) and virtual influencers.

Our research also provides actionable insights for marketers. Building on the finding that virtual influencers can be effective by leveraging their perceived uniqueness, marketers can enhance campaign impact by highlighting their unique attributes, and continually emphasizing their evolving customization and flexibility. Tailored campaigns that positively resonate with specific market segments can further capitalize on this uniqueness. Moreover, to build the trust and engagement essential for facilitating a personal connection between consumers and virtual influencers, content that reflects audience values and interests can used to foster a deep identification between the audience and influencer.

2. Conceptual development

Virtual influencers are computer-generated characters that use social platforms to share content and interact with users, aiming to significantly influence consumers (Mouritzen et al., 2023). While various types of virtual influencers exist, including human-like as well as cartoon or animal-like characters, we specifically concentrate on hyper-realistic virtual influencers, as categorized by Mouritzen et al. (2023). These virtual influencers so closely resemble humans that distinguishing them from real-life influencers can be challenging. Their content often revolves around their personal lives, often in real-world physical settings, thereby blurring the line between the virtual and real worlds.

Next, we draw upon various theories and literature streams to develop our conceptual framework. We begin by highlighting the significance of the *Computers Are Social Actors (CASA)* theory and discuss insights from anthropomorphism (Gambino et al., 2020; Reeves & Nass, 1996) to aid our understanding of consumer responses to hyper-realistic versus cartoon-like virtual influencers. Then, we explore the literature on consumers' differential reactions to hyper-realistic virtual versus human influencers. Specifically, we first focus on the well-accepted insight that hyper-realistic virtual influencers often yield more negative perceptions and outcomes than humans (e.g., Li et al., 2023; Stein et al., 2022; Wan & Jiang, 2023). The *uncanny valley* theory (Mori, 1970) provides further insight on this negative effect and supports the inference that negative reactions to hyper-realistic virtual influencers ensue from a lower level of perceived trustworthiness. Subsequently, we shift our focus to hyper-realistic virtual influencers' positive perceptions, which are the primary focus of this research. We infer that these positive perceptions arise from viewing these influencers as a unique type of endorser. To illuminate this perspective, we explore the concept of *individuation* (Maslach et al., 1985), which elucidates how consumers perceive virtual influencers as unique individuals, thereby influencing positive outcomes. Consequently, we anticipate the validity of this proposition, particularly when consumers identify with the endorser, drawing upon insights from *social identity* theory (Tajfel & Turner, 2004).

2.1. Hyper-realistic versus cartoon-like virtual influencers

Extant research on virtual influencers focuses on comparing the effectiveness of hyper-realistic virtual and cartoon-like influencers (Arsenyan & Mirowska, 2021; de Brito Silva et al., 2022; Xie-Carson et al., 2023; Yang et al., 2023). Studies reveal that compared with cartoon-like influencers, hyper-realistic virtual influencers may elicit perceptions which are more similar to those of humans (Xie-Carson et al., 2023; Yang et al., 2023). This underscores the impact of anthropomorphism in shaping virtual influencers' positive perceptions. Hence, this concept aligns with the CASA and anthropomorphism theories. According to these theories, when computers display human-like characteristics, such as voice, appearance, or emotional expressions, they can evoke emotional responses in people (Gambino et al., 2020; Reeves & Nass, 1996). Then, given their striking resemblance to humans, hyper-realistic virtual influencers may elicit similar emotional and perceptual responses, thereby effectively making them akin to their human counterparts.

2.2. Hyper-realistic virtual versus human influencers

Hyper-realistic virtual influencers closely resemble humans in terms of their physical appearance and behavior. However, they are not human beings. This distinction can induce both positive and negative perceptions depending on whether consumers focus on the aspects where virtual influencers fall short compared to humans (i.e., seen as less effective brand endorsers) or qualities that virtual influencers possess in addition to humans (i.e., seen as superior brand endorsers). Essentially, people's opinions about virtual influencers can vary based on which aspects people prioritize or consider when evaluating these influencers.

2.2.1. Negative perceptions and outcomes of hyper-realistic virtual (versus human) influencers

To uncover these perceptions and their effectiveness as brand endorsers, research on hyper-realistic virtual influencers has focused on comparing their effectiveness versus human counterparts. Specifically, studies investigate whether, why, and when virtual influencers can potentially replace human influencers (e.g., Franke et al., 2022; Li et al., 2023; Sands et al., 2022; Stein et al., 2022; Wan & Jiang, 2023).

Most studies tend to highlight the downsides of utilizing virtual influencers in advertising campaigns and explore ways to mitigate them. For instance, virtual influencers are considered to be less effective than human influencers due to their limited sensory capabilities, lower credibility, and higher social distance. However, this is not the case when the advertisements successfully reduce consumers' reliance on sensory cues (Li et al., 2023), promote technical (versus cosmetic) products (Franke et al., 2022), or when the target consumers are high in need for uniqueness (Sands et al., 2022).

Furthermore, virtual influencers encounter significant challenges in fostering parasocial relationships because consumers perceive them to lack human-like mental attributes and similarity (Stein et al., 2022). Virtual influencers are also seen as less capable of prompting meaningful dialogue, and less warm and trustworthy than their human counterparts (Wan & Jiang, 2023). Multiple studies consistently underscore that consumers tend to have less trust in virtual influencers regarding product and brand endorsements (Byun & Ahn, 2023; Hofeditz et al., 2022; Sands et al., 2022; Wan & Jiang, 2023). Specifically, in a systematic review comparing the similarities and differences between human and virtual influencers, Byun and Ahn (2023) argue that people tend to trust the former less than the latter. Similarly, Hofeditz et al. (2022), Sands et al. (2022), and Wan and Jiang (2023) find that perceived trust is consistently rated higher for human rather than virtual influencers.

The uncanny valley theory can help explain virtual influencers' lower trustworthiness than humans (Mori, 1970). Specifically, while making things appear more human often generates positive reactions, excessive human-like features can trigger negative responses. As a robot becomes increasingly human-like but just falls short of perfect resemblance, it may rather evoke negative emotions than positive ones (Mori, 1970). This discomfort can occur when interacting with entities like humanoid robots, animated characters, or virtual influencers that closely mimic humans but have subtle differences (Arsenyan &

Mirowska, 2021; Block & Lovegrove, 2021; Kim et al., 2019; Lou et al., 2022; Tinwell et al., 2013), thereby leading to unfavorable responses (Franke et al., 2022). Consequently, this issue may impact trustworthiness, a key factor in brand endorsement effectiveness (Ohanian, 1990). Therefore, virtual influencers' trustworthiness compared to their human counterparts is crucial. Distrust in virtual influencers may hinder favorable consumer responses.

2.2.2. *Positive perceptions and outcomes of hyper-realistic virtual (versus human) influencers*

Contrary to these prevailing negative perceptions and outcomes, hyper-realistic virtual influencers' extensive use in advertising campaigns implies that they may have qualities that resonate with consumers. This can potentially make them more effective than human influencers in specific contexts. Indeed, virtual influencers possess unique and distinguishing traits that distinguish them from their human counterparts. Therefore, we suggest that these distinctive characteristics can create a sense of uniqueness in consumers' perceptions, thereby enhancing virtual influencers' effectiveness in endorsing products and brands compared with human influencers.

We refer to perceived uniqueness as "individuation," a concept that measures the extent to which something, a behavior, or a person is perceived as specific and unique in contrast to similar entities (Maslach et al., 1985). Thus, we define perceived uniqueness as the exclusive characteristics that differentiate virtual influencers from their human counterparts. A primary characteristic contributing to their individuation or perceived uniqueness is their novelty. This novelty arises from people's fascination with influencers which use cutting-edge technology. This fascination motivates people to follow these influencers and brings a sense of novelty to the advertising campaigns they are involved in (Lou et al., 2022; Franke et al., 2022). Another distinctive aspect is the high degree of customization that virtual influencers offer. They can be tailored in their appearance, stories, and behavior, aligning precisely with the interests of specific target audiences (Conti et al., 2022).

Furthermore, virtual influencers exhibit an impressive level of flexibility that surpasses human limitations, such as fatigue, hunger, or the need to be available at all times. This flexibility enables them to maintain a constant online presence (Byun & Ahn, 2023; Conti et al., 2022). Lastly, they excel at creating a mixed reality by seamlessly blending virtual and real-world elements into their content. This allows them to appear at physical events or collaborate with human celebrities, creating an interactive and immersive consumer experience (Block & Lovegrove, 2021; Ham et al., 2023; Lou et al., 2022). These unique characteristics are the primary drivers of consumer attention toward virtual influencers (de Brito Silva et al., 2022; Franke et al., 2022; Lou et al., 2022).

In summary, virtual influencers possess distinct characteristics that distinguish them from human influencers. We contend that these traits combine to create a perception of uniqueness among consumers. This perceived uniqueness, in turn, is a key factor driving the persuasive impact of virtual influencers as brand endorsers. According to Lee and Eastin (2021), for more "traditional" endorsers such as celebrity endorsers, characteristics like trustworthiness or attractiveness are paramount; meanwhile, social media influencers heavily rely on their perceived uniqueness to cultivate an aura of authenticity. This authenticity fosters a sense of identification among consumers, leading them to follow and purchase endorsed products. This difference in the origin of social media influencers' authenticity and persuasiveness compared to celebrity endorsers likely stems from a different approach to the practice of creating endorsements. Where celebrity endorsers are creating specific messages designated to brand recommendations, influencers largely integrate brands in their everyday lives and report on this in their posts on social media (Lee & Eastin, 2021).

Consequently, we propose that virtual influencers' perceived uniqueness compared with their human counterparts generates interest as well as strengthens intentions to follow and make purchases based on their recommendations. This proposition is analogous to findings reported in extant research. For instance, perceived uniqueness plays a significant role for individuals regarded as opinion leaders on Instagram. It influences various consumer behaviors, such as interactions, recommendations, intentions to follow the influencer, and intentions to purchase the recommended product (Casaló et al., 2020; Fakhreddin & Foroudi, 2022). Likewise, the perceived uniqueness of an influencer's content can substantially affect followers' purchase intentions (Farivar et al., 2023; Farivar & Wang, 2021).

Building upon these findings, we argue that compared with human influencers, virtual influencers' perceived uniqueness can foster positive behavioral outcomes. According, we propose the following hypothesis:

> **H1:** Virtual influencers, as opposed to human influencers, positively impact perceived uniqueness, leading to an increase in both (a) purchase intention and (b) intention to follow.

Nevertheless, we do acknowledge that hyper-realistic virtual (versus human) influencers may simultaneously generate negative perceptions, such as trustworthiness issues. Accordingly, we argue that trustworthiness can function as a parallel mediator that competes with perceived uniqueness. This leads to our second hypothesis:

> **H2:** Virtual influencers, as opposed to human influencers, lower perceived trustworthiness, resulting in reduced (a) purchase intention and (b) intention to follow.

Considering our previous discussion, hyper-realistic virtual influencers can trigger both positive and negative perceptions. These perceptions, in turn, can either positively or negatively influence behavioral outcomes. We propose that the extent to which consumers identify with the influencer plays a pivotal role in whether the total outcome is positive or negative.

In this context, "identification" refers to the perceived similarity between individuals and influencers (Hoffner & Buchanan, 2005). This concept draws from social identity theory, which explains how individuals form their own identity by aligning with specific social groups (Tajfel & Turner, 2004). On social media, influencers play a pivotal role in inspiring individuals to shape their identities within digital social groups (Carlson et al., 2020). This phenomenon is closely linked to the concept of parasocial relationships, where identification often emerges as an outcome. Repeated interactions with influencers can foster feelings of intimacy, perceived friendship, and a sense of identification with the influencer (Chung & Cho, 2017; Horton & Wohl, 1956). This process, where people establish connections and identify with others, is applicable in influencer marketing through consumer-endorser identification. Consumer-endorser identification involves matching oneself with an endorser. It is a significant factor influencing brand endorsement effectiveness (Choi & Rifon, 2012; Xu & Pratt, 2018).

Essentially, consumer-endorser identification refers to the feeling of similarity that individuals experience with endorsers who share demographic characteristics, like gender, race and age, or personal traits, such as personality, behavior patterns, or life experiences (Hoffner, 1996). For example, Schouten et al. (2019) suggest that consumers often relate more to influencers than to celebrities. Influencers are perceived as relatable and approachable, while celebrities often seem distant due to their glamorous and less relatable lifestyles. This sense of similarity with influencers makes consumers more open to product claims from influencers than celebrities. Furthermore, numerous studies consistently show that stronger consumer identification with endorsers significantly influences consumers' attitudes and intentions toward products or services (Carlson et al., 2020; Hu et al., 2020; Janssen et al., 2022; Lou & Yuan, 2019).

In a similar vein, we propose that consumer-endorser identification significantly influences the effectiveness of endorsements by virtual compared to human influencers. Specifically, we argue that while virtual influencers are often seen as more unique, they may also face

challenges related to trustworthiness. The strength of consumer-endorser identification determines whether these positive perceptions of uniqueness or less favorable perceptions of trustworthiness ultimately result in positive or negative responses. When a consumer does not strongly identify with an influencer, higher perceived uniqueness may not be valuable for them. In such cases, uniqueness may not strongly impact their behavioral responses. However, when consumers strongly identify with the influencer, high perceived uniqueness can lead to even more favorable behavioral outcomes. The same principle applies to perceptions of trustworthiness. If consumers do not have a strong identification with the influencer, a less favorable perception of trustworthiness may not greatly affect their responses. However, when consumers strongly identify with the influencer, whether virtual or human, high perceived trustworthiness can lead to even more favorable behavioral outcomes. Essentially, trust becomes a crucial element when consumers closely relate to the influencer. Accordingly, we hypothesize the following:

H3: The positive impact of uniqueness on (a) purchase intention and (b) intention to follow strengthens (weakens) when identification with the influencer is high (low).
H4: The positive impact of trustworthiness on (a) purchase intention and (b) intention to follow strengthens (weakens) when identification with the influencer is high (low).

Fig. 1 shows our conceptual framework:

3. Overview of studies

Employing four studies, we investigated hyper-realistic virtual influencers' impact compared to human influencers on perceived uniqueness, and in turn, on an increase in purchase intention and intention to follow. Studies 1a and 1b examine hypothesis H1, which suggests that the perception of uniqueness positively influences virtual influencers' brand endorsement compared to human influencers. In Study 2, we simultaneously tested hypotheses H1 and H2, examining the competing mediation roles of perceived uniqueness and trustworthiness. Finally, in Study 3, we examined hypothesis H3 and explored the moderated-mediation effects. Specifically, we investigated how consumer-endorser identification moderates the impact of uniqueness and trustworthiness on behavioral outcomes. These studies enabled us to test our full conceptual model.

3.1. Studies 1a-1b

In Studies 1a and 1b, we investigated how consumer perceptions of uniqueness drive the contrasting impact of virtual versus human influencers on brand endorsement effectiveness specifically in terms of purchase intention and intention to follow (H1). In Study 1a, we examined this effect using a fast-moving consumer good (FMCG), namely an energy bar. In Study 1b, we replicated the effect with a durable good, namely a luxury handbag.

3.1.1. Method

Design and stimuli. For both studies, we employed a between-participant design to examine the impact of influencer type (human versus virtual). Participants were randomly assigned to one of two conditions. In each condition, participants were presented with both text and visual examples of an influencer endorsing a product. We manipulated the type of the influencer described in the text (virtual versus human) by altering the textual description while keeping the visual stimuli the same. In the text of Study 1a, we first explained the different types of influencers, such as those emerging from social media (human and virtual influencers). Then, we provided an example of either a human or virtual influencer, along with additional information about the influencer from each condition. In the text of Study 1b, we only specified whether the influencer was virtual or human. In the virtual influencer condition, we explained what virtual influencers are and then provided additional information about the specific influencer featured in this study. The additional information included the influencer's name, years of activity, Instagram popularity, and the brands typically endorsed by the influencer. Participants were then informed that the influencer had recently posted an endorsement for a brand and a visual example (an Instagram post) was presented.

The influencer and post were sourced from Instagram. In Study 1a, we featured a virtual influencer named Rozy from South Korea. Meanwhile, in Study 1b, we used a human influencer named Emma from the United States. To enhance the ecological validity in our scenarios, we included a photo of both influencers endorsing real products. In Study 1a, the promoted product was an energy bar and we used a fictitious brand name. In Study 1b, the promoted product was a Louis Vuitton bag. We chose Instagram as the platform for these scenarios since virtual influencers are primarily active on this platform. To clarify, in both conditions of each study, we used the exact same picture of the influencer and simply labeled the virtual influencers differently (humans versus virtual) in the text. For a comprehensive description of the scenarios and stimuli used in Studies 1a and 1b, please see Appendix A.

Procedure. Participants were initially informed that they would be reading a section of a magazine article and would later be asked questions about it. They were instructed to read the content with attention. Subsequently, they were randomly shown the magazine article, accompanied by a visual example of the influencer endorsing a product. Finally, they were tasked with answering the questions.

Measures. We assessed purchase intention using a single-item question adapted from Schouten et al. (2019): "Next time you are looking to purchase an energy bar/bag, would you consider buying this product endorsed by Rozy/Emma?". Participants responded on a seven-point scale ranging from 1 (Extremely unlikely) to 7 (Extremely likely). For intention to follow, we employed a single-item question based on Sands et al. (2022): "How likely would you be to follow Rozy/Emma on social

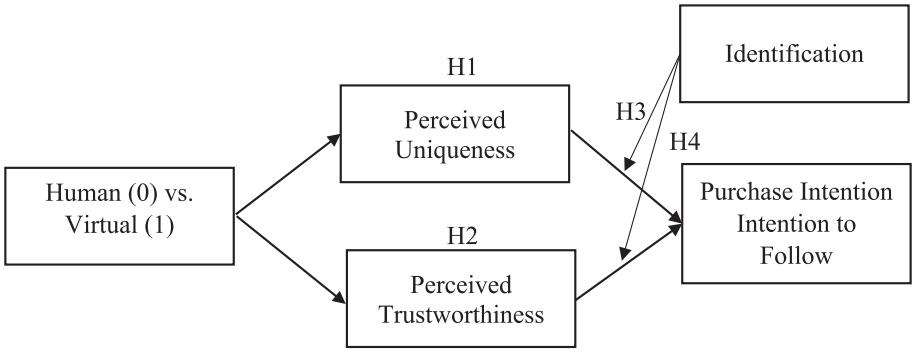


Fig. 1. Conceptual framework.

media?". This was also rated on a seven-point scale, ranging from 1 (Extremely unlikely) to 7 (Extremely likely). To measure uniqueness, we used a scale adapted from Vigneron and Johnson (2004). Participants rated the influencer's perceived exclusivity, rarity, uncommonness, uniqueness, and unusualness on a seven-point Likert scale, ranging from 1 (Not at all) to 7 (Very much) ($\alpha_{Study1a} = 0.86$; $\alpha_{Study1b} = 0.89$).

Subsequently, we included two control questions. However, we did not include traditional manipulation check questions that asked participants, for example, how much they perceived the influencer to be virtual or human. This is because we manipulated the text and not the visual aspect. In the visuals, they looked the same, and people could not differentiate between them. We needed to prompt a question regarding the text we were manipulating. Therefore, we decided to employ an alternative method to assess and control the impact of our manipulation. Similar to Franke et al. (2022), we included control or attention-check questions to verify if participants accurately recalled the type of influencer featured in the text/post, distinguishing between human and virtual influencers, and whether participants recognized the influencer mentioned in the figure. This was to ensure that our results were unbiased. Some participants may have confused the human influencer with the virtual one or vice versa, potentially causing them to answer the questions based on a mistaken identity of the influencer. Additionally, some participants may have been familiar with the influencer due to prior exposure, which could have led to pre-existing perceptions about them. Participants who either did not recall or who recognized the influencer were excluded from the analysis. Finally, after participants completed demographic questions, including those on age and gender, they were thanked for their participation.

3.1.2. Results

Influencer recall and recognition check. In Study 1a, we excluded 20 participants from the analysis because they failed the control questions mentioned above, resulting in a final sample of 120 participants from the United States recruited through Prolific (70 women; *Mage* = 34.99, *SD* = 12.62). In Study 1b, we excluded 67 participants from the analysis who also failed the control questions, leaving us with a sample of 219 participants from the United States also recruited through Prolific (105 women; *Mage* = 36.79, *SD* = 12.08).

Mediation by perceived uniqueness. Two mediation analyses were conducted in each study (1a-1b) using PROCESS (Model 4 of the Process Macro) with 10,000 bootstrapping samples (Hayes, 2017). We included influencer type as the independent variable, purchase intention and intention to follow as the dependent variables, and perceived uniqueness as the mediator.

The results of Study 1a revealed that virtual (versus human) influencers significantly predicted perceived uniqueness (a = 0.74, SE = 0.25, $p = 0.004$). Perceived uniqueness, in turn, significantly predicted both purchase intention (b = 0.33, SE = 0.10, $p = 0.001$) and intention to follow (b = 0.34, SE = 0.09, $p = 0.001$). While the total effect was not significant (c = −0.45, SE = 0.29, 95 % CI [−1.04, 0.13]), the remaining direct effect was significant for purchase intention ($c' = -0.70$, SE = 0.29, 95 % CI [−1.28, −0.11]). However, both the total and directly effects were not significant for the intention to follow (c = −0.28, SE = 0.28, 95 % CI [−0.84, 0.27]; $c' = -0.53$, SE = 0.28, 95 % CI [−1.08, 0.02]). Still, as expected, the indirect effect was significant for both dependent variables (ab = 0.23, SE = 0.11, 95 % CI [0.04, 0.50]; ab = 0.24, SE = 0.11, 95 % CI [0.05, 0.50]). Therefore, hypothesis H1 was supported for both purchase intention and intention to follow.

In study 1b, the analysis revealed that virtual (versus human) influencers significantly predicted perceived uniqueness (a = 1.00, SE = 0.19, $p < 0.001$). Perceived uniqueness significantly predicted purchase intention (b = 0.27, SE = 0.07, $p < 0.001$) and intention to follow (b = 0.43, SE = 0.06, $p < 0.001$). Similar to Study 1a, the total and remaining direct effects were not significant for purchase intention (c = 0.08, SE = 0.22, 95 % CI [−0.35, 0.52]; $c' = -0.19$, SE = 0.23, 95 % CI [−0.64, 0.25]) and intention to follow (c = 0.07, SE = 0.20, 95 % CI [−0.33, 0.47]; $c' = -0.36$, SE = 0.19, 95 % CI [−0.75, 0.02]). As anticipated, the indirect effect remained significant for both dependent variables (ab = 0.27, SE = 0.10, 95 % CI [0.09, 0.51]; ab = 0.43, SE = 0.12, 95 % CI [0.22, 0.69]). These findings are similar to those observed in Study 1a, albeit with a different type of product. For results of the independent *t*-test, which examined the direct effect of the independent variables (influencer type: human versus virtual) on the dependent variables (purchase intention and intention to follow) without the mediator, see Appendix B.

3.1.3. Discussion

The findings from Studies 1a and 1b underscore the positive impact of perceiving virtual influencers as unique, in contrast to human influencers, on their effectiveness in endorsing products. This supports the idea that recognizing the unique traits of virtual influencers not only captures consumers' attention (de Brito Silva et al., 2022; Franke et al., 2022; Lou et al., 2022) but also influences their intention to purchase the promoted product/brand and to follow the influencer. The concept of uniqueness has previously been shown to drive favorable behaviors among opinion leaders on Instagram (Casaló et al., 2020). The results of both studies suggest that this perception plays a similar role when studying virtual influencers. Furthermore, as demonstrated in our studies, the intention to purchase products promoted by virtual influencers compared to human counterparts is applicable to FMCG like energy bars, as well as durable products like luxury handbags.

Unsurprisingly, compared with human influencers, virtual influencers' direct impact on purchase intention and the intention to follow was both negative and statistically non-significant. Research on virtual influencers consistently demonstrates that consumers tend to place less trust in virtual influencers than in humans, often preferring endorsements from humans over virtual influencers (Byun & Ahn, 2023; Hofeditz et al., 2022; Sands et al., 2022; Wan & Jiang, 2023). Therefore, our next study aimed to broaden our understanding of consumer perceptions toward virtual influencers by exploring trustworthiness as an additional mediating factor alongside perceived uniqueness.

3.2. Study 2

In Study 2, we explored the parallel mediation of uniqueness and trustworthiness, and whether trustworthiness acts as a competing mediator, diminishing the effectiveness of virtual influencers compared to human influencers in brand endorsement. Essentially, we tested for H1 and H2. Additionally, unlike studies which focused on the effectiveness of female virtual influencers, this study introduced a male influencer into the analysis.

3.2.1. Method

Design and stimuli. We utilized a between-participant design based on influencer type (human versus virtual). Participants were randomly assigned to one of two conditions. The scenarios in each condition followed the same format as in Study 1b. Here, the featured influencer was a model sourced from an open-source image website. We referred to him as Brad. Furthermore, the endorsement post was created using an image editing software. The product endorsed by the influencer was a shirt. For a comprehensive description of the scenario and stimuli, please refer to Appendix A.

Procedure. Consistent with our previous studies, participants were initially informed that they would read a magazine article. They were then randomly shown the magazine article along with a visual example of the influencer endorsing a product, following which they answered questions.

Measures. We used the same items as in our previous studies to measure the dependent variables (purchase intention and intention to follow), and the same scale to measure the mediator perceived uniqueness ($\alpha = 0.91$). Additionally, we incorporated the trustworthiness scale from Ohanian (1990). It assessed the extent to which

participants perceived the influencer Brad as undependable to dependable, dishonest to honest, unreliable to reliable, insincere to sincere, and untrustworthy to trustworthy, rated on a seven-point bipolar scale ($\alpha = 0.93$). Furthermore, the control and demographic questions remained consistent with those used in our previous studies.

3.2.2. Results

Influencer recall and recognition check. We excluded 17 participants from the analysis due to failing control questions, resulting in a final sample of 175 participants from the United States recruited through Prolific (78 Women, *Mage* = 35.66, SD = 12.76).

Parallel-Mediation. We conducted two parallel mediation analyses employing PROCESS (Model 4 of the Process Macro) with 10,000 bootstrapping samples (Hayes, 2017). The independent variable was influencer type, the dependent variables were purchase intention and intention to follow, and perceived uniqueness and trustworthiness served as the two mediators.

The analysis revealed that virtual (versus human) influencers significantly predicted perceived uniqueness ($a = 1.25$, SE = 0.22, $p < 0.001$) and trustworthiness ($a = -0.36$, SE = 0.21, $p = 0.098$). Perceived uniqueness significantly predicted purchase intention ($b = 0.19$, SE = 0.06, $p = 0.004$) and intention to follow ($b = 0.16$, SE = 0.06, $p = 0.014$). The total effect was not significant for both purchase intention ($c = 0.14$, SE = 0.23, 95 % CI [−0.25, 0.53]) and intention to follow ($c = 0.14$, SE = 0.23, 95 % CI [−0.23, 0.53]). The remaining direct effect was also not significant for both dependent variables ($c' = 0.14$, SE = 0.23, 95 % CI [−0.33, 0.61]; $c' = 0.14$, SE = 0.21, 95 % CI [−0.33, 0.49]). However, as predicted, the mediation effect of perceived uniqueness was positive and significant (ab = 0.24, SE = 0.10, 95 % CI [0.07, 0.47]). Meanwhile, the mediation effect of trustworthiness was negative and marginally significant (ab = −0.20, SE = 0.11, 90 % CI [−0.39, −0.00]). Thus, H2 regarding purchase intention was supported, considering a 95 % CI for the indirect effect of the mediator uniqueness and 90 % CI for the indirect effect of the mediator trustworthiness.

The mediation effect of uniqueness was positive and significant (ab = 0.24, SE = 0.10, 95 % CI [0.07, 0.47]) and of trustworthiness was negative and marginally significant for the intention to follow (ab = −0.13, SE = 0.08, 95 % CI [−0.26, −0.00]). Therefore, H2 regarding intention to follow was also supported, considering a 95 % CI for the indirect effect of the mediator uniqueness and 90 % CI for the indirect effect of the mediator trustworthiness.

3.2.3. Discussion

The results from Study 2 revealed that consumers have mixed feelings and perceptions, encompassing both positive and negative aspects, towards virtual influencers. In particular, trustworthiness is a negative perception that coexists with the positive perception of uniqueness. These findings are consistent with studies which show that consumers tend to trust virtual influencers less than their human counterparts (Byun & Ahn, 2023; Hofeditz et al., 2022; Sands et al., 2022; Wan & Jiang, 2023). However, note that virtual influencers do not exclusively evoke negative perceptions. Our two previous studies demonstrated that when consumers perceive virtual influencers to be unique, consumers have positive responses towards the products and brands they endorse.

Therefore, by highlighting the interplay between uniqueness and trustworthiness as conflicting factors, we underscore the intricate and diverse feelings and perceptions that consumers hold towards virtual influencers. To delve deeper into this phenomenon, our next study investigated the significant role of consumer-endorser identification as a moderating factor. Specifically, it examined whether positive or negative perceptions of virtual influencers, compared with those of human influencers, on behavioral outcomes are influenced by the degree to which consumers can identify with the influencer.

3.3. Study 3

In Study 3, we examine the moderated-mediation effect of perceived uniqueness and consumer-endorser identification (H3), and trustworthiness and consumer-endorser identification (H4) on behavioral outcomes.

3.3.1. Method

Design and stimuli. We employed a between-participant design by influencer type (human versus virtual). The participants were randomly assigned to one of the two conditions. The scenarios in each condition followed the same format as in Study 1a. The influencer featured in the study, Daisy, is a virtual influencer from Italy. She endorses a Gucci bag. For a comprehensive description of the scenario and stimuli, please see Appendix A.

Procedure. In line with our prior research, participants were first informed that they would be reading a magazine article. Subsequently, they were randomly presented with both the magazine article and a visual representation of the influencer endorsing a product before proceeding to answer questions.

Measures. We used the same measurement scales as in our previous studies to assess the dependent variables: purchase intention and intention to follow, and the mediators, perceived uniqueness ($\alpha = 0.90$) and trustworthiness ($\alpha = 0.95$). Additionally, to measure the moderator, we introduced the Consumer-Endorser Identification Scale from Hoffner and Buchanan (2005). This scale assessed participants' extent of agreement with the following statements on a seven-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree): "Daisy is the type of person I want to be like myself"; "Sometimes I wish I could be more like Daisy"; "Daisy is someone I would like to imitate"; and "I would like to do the kind of things Daisy does" ($\alpha = 0.94$). Furthermore, we included the same control and demographic questions as in our previous studies.

3.3.2. Results

Influencer recall and recognition check. We excluded 62 participants from the analysis because they failed to answer the control questions, resulting in a final sample of 342 participants from the United States recruited through Prolific (*Mage* = 42.23, *SD* = 14.06).

Moderated-mediation results. We conducted two parallel moderated-mediation analyses using PROCESS (Model 14 of the Process Macro) with 10,000 bootstrapping samples (Hayes, 2017). We treated influencer type as the independent variable, while purchase intention and intention to follow served as the dependent variables. We examined perceived uniqueness and trustworthiness as the two mediators. Meanwhile, we introduced consumer-endorser identification as the moderator for the "b paths," which represented the impact of perceived uniqueness and trustworthiness on behavioral outcomes.

The analysis revealed that virtual (versus human) influencers significantly affected perceived uniqueness ($a = 1.03$, SE = 0.16, $p < 0.001$) and trustworthiness ($a = -0.64$, SE = 0.16, $p < 0.001$), albeit in opposite directions. The interaction of perceived uniqueness and consumer-endorser identification was significant for purchase intention ($b = 0.07$, SE = 0.03, $p = 0.017$) and marginally significant for intention to follow ($b = 0.05$, SE = 0.02, $p = 0.050$). The interaction of trustworthiness and consumer-endorser identification was significant for both purchase ($b = 0.12$, SE = 0.03, $p = 0.001$) and intention to follow ($b = 0.06$, SE = 0.02, $p = 0.025$).

The remaining direct effect was not significant for purchase intention ($c' = 0.02$, SE = 0.13, 95 % CI [−0.23, 0.28]), but was significant for intention to follow ($c' = -0.27$, SE = 0.12, 95 % CI [−0.51, −0.03]). Importantly, as predicted, the index of moderated-mediation of perceived uniqueness and consumer-endorser identification on purchase intention was positive and significant (ab = 0.07, SE = 0.03, 95 % CI [0.01, 0.15]. Furthermore, the index of moderated-mediation of trustworthiness and consumer-endorser identification was negative and

significant (ab = −0.08, SE = 0.03, 95 % CI [−0.15, −0.02]).

With a spotlight analysis, we decomposed the interaction term to determine the effect of perceived uniqueness and perceived trustworthiness on purchase intention at different levels of consumer-endorser identification. First, the effect of perceived uniqueness on purchase intention was significant when consumers had a high (1 SD above the mean: 3.73; B = 0.28, SE = 0.06, t = 4.21, $p < 0.001$) or moderate (mean value: 2.25, B = 0.17, SE = 0.04, t = 4.01, $p = 0.001$) level of identification with the influencer. However, it was not significant when consumers had a low level of identification (1 SD below the mean: 1.00; B = 0.08, SE = 0.05, t = 1.61, $p = 0.106$). In addition, a floodlight analysis showed that the Johnson-Neyman point had a value of 1.17 for identification (in the model considering purchase intention), implying that the effect of uniqueness on purchase intention was positive and significant only for identification levels above this point. Therefore, H3 on purchase intention was supported. Fig. 2 shows an illustration of the moderated-mediation effect of consumer-endorser identification and the mediator perceived uniqueness on purchase intention.

Similarly, the effect of perceived trustworthiness on purchase intention was significant when consumers had a high (1 SD above the mean: 3.73; B = 0.38, SE = 0.07, t = 5.16, $p < 0.001$) or moderate (mean value: 2.25, B = 0.20, SE = 0.04, t = 4.05, $p = 0.001$) level of identification with the influencer. However, it was not significant when consumers had a low level of identification (1 SD below the mean: 1.00; B = 0.04, SE = 0.05, t = 0.79, $p = 0.427$). In addition, a floodlight analysis showed that the Johnson-Neyman point had a value of 1.44 for identification (in the model considering purchase intention), implying that the effect of uniqueness on purchase intention was positive and significant only for identification levels above this point. Therefore, H4 on purchase intention was supported. Fig. 3 shows an illustration of the moderated-mediation effect of consumer-endorser identification and the mediator trustworthiness on purchase intention.

For the intention to follow, the moderated-mediation index did not demonstrate statistical significance for perceived uniqueness and identification (ab = 0.05, SE = 0.03, 95 % CI [−0.00, 0.12]), or trustworthiness and identification (ab = −0.04, SE = 0.02, 95 % CI [−0.09, 0.00]) using the conventional 95 % CI.

Hence, hypotheses H3 and H4 cannot be supported for the intention to follow based on the conventional level of significance. Still, note that in both cases, the intervals included 0 as their upper/lower limits, suggesting borderline significance. Indeed, at an alpha level of 0.90, the results were significant, suggesting marginal significance for both perceived uniqueness and identification (ab = 0.05, SE = 0.03, 90 % CI [0.00, 0.11]), and trustworthiness and identification (ab = −0.04, SE = 0.02, 90 % CI [−0.08, −0.00]).

When we decomposed the interaction term to assess the impact of perceived uniqueness and perceived trustworthiness on the intention to follow at different levels of consumer-endorser identification, we found

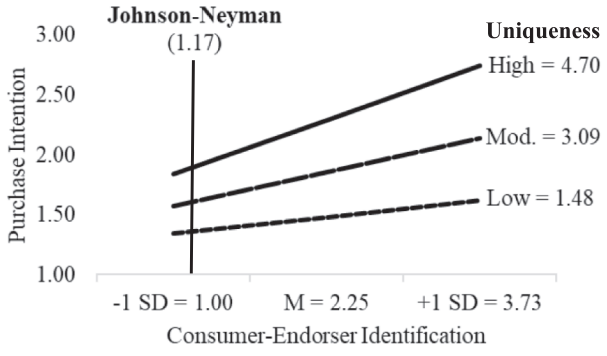


Fig. 2. Illustration of the moderated-mediation effect of consumer-endorser identification and perceived uniqueness on purchase intention.

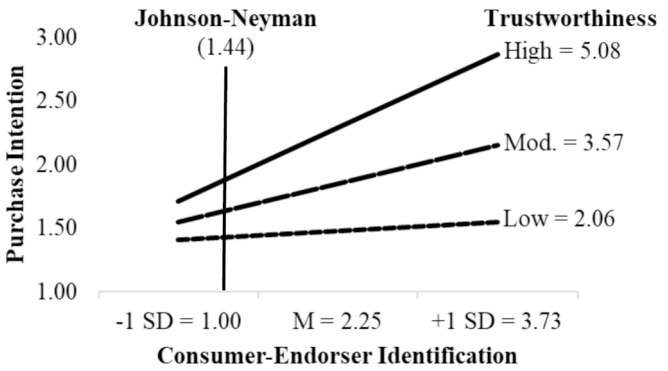


Fig. 3. Illustration of the moderated-mediation effect of consumer-endorser identification and perceived trustworthiness on purchase intention.

no difference in the impact of perceived uniqueness (1 SD above the mean: 3.72; B = 0.27, SE = 0.06, t = 4.43, $p < 0.001$; mean value: 2.25, B = 0.19, SE = 0.04, t = 4.78, $p < 0.001$; 1 SD below the mean: 1.00; B = 0.12, SE = 0.04, t = 2.56, $p = 0.010$). However, the pattern of the results was similar to that for purchase intention: decreasing impact of perceived uniqueness on purchase intention/intention to follow with decreasing consumer-endorser identification, as indicated by the B-coefficients. Similarly, we found no difference for trustworthiness (1 SD above the mean: 3.72; B = 0.32, SE = 0.06, t = 4.75, p < 0.001; mean value: 2.25, B = 0.23, SE = 0.04, t = 5.05, p < 0.001; 1 SD below the mean: 1.00; B = 0.15, SE = 0.05, t = 2.89, p = 0.004). Again, the pattern of the results followed that observed for purchase intention: decreasing B-coefficients for decreasing consumer-endorser identification levels. Hence, although H3 and H4 cannot be supported for intention to follow considering conventional levels of significance, the results do imply that perceived uniqueness and perceived trustworthiness have larger impacts on intention to follow when consumers can identify with the virtual influencer.

3.3.3. *Discussion*

In Study 3, we demonstrated that influencers' perceived uniqueness and trustworthiness positively affected consumers' purchase intention, but only when consumers identify with the influencer. When consumers could not relate to the influencer, the impact of perceived uniqueness and trustworthiness on purchase intentions became weaker, and neither factor significantly influenced purchase intentions. However, while these effects were observed for purchase intentions, they were less pronounced and not significant on conventional levels for intention to follow.

Identification's crucial role in establishing down-stream effects has consistently emerged in the human influencer marketing literature (Carlson et al., 2020; Hu et al., 2020; Janssen et al., 2022; Lou & Yuan, 2019; Schouten et al., 2019). Moreover, it holds true for virtual influencers as well. Therefore, virtual influencers' perceived uniqueness can be expected to affect consumers' purchase intention of the endorsed products only when consumers experience self-influencer identification, just as it applies to trustworthiness perceptions with the human counterparts.

The difference in how consumer-endorser identification affects uniqueness and trustworthiness concerning purchase intention, as opposed to intention to follow, can be attributed to the commitment required for these actions. For purchase intention, a strong sense of trust in the person recommending the product or brand is often essential. Therefore, identification with the influencer is necessary. Meanwhile, the intention to follow an influencer on social media is a relatively low-commitment action, often driven by curiosity or enjoyment of their content, rather than a strong intention to actively engage with their endorsed products. This aligns with research that emphasizes curiosity as a factor attracting attention to virtual influencers (de Brito Silva et al.,

2022; Franke et al., 2022; Lou et al., 2022). Hence, even if consumers perceive an influencer to be unique or trustworthy, consumers' level of identification may not significantly impact their intention to follow, as the decision to follow can be influenced by factors like curiosity. However, regarding purchasing a product, a higher (versus lower) level of identification with the influencer is likely required to drive the action.

4. General discussion

This research explored brand endorsements' effectiveness by virtual influencers compared to human influencers. It introduced a positive characteristic that may potentially offset virtual influencers' negative characteristics reported in prior literature: perceived uniqueness. We found that perceived uniqueness is indeed an important factor which positively influences consumer responses. Similarly, studies have identified perceived uniqueness as a factor influencing various consumer behaviors (e.g., Casaló et al., 2020; Farivar et al., 2023; Lindmoser et al., 2022). Furthermore, we demonstrated that perceived uniqueness influences purchase intention and the intention to follow when virtual influencers endorse products and brands in advertising (Studies 1a and 1b). Additionally, these findings help explain previous research aimed at mitigating virtual influencers' negative impact. Specifically, our results contribute to a better understanding of why consumers with a high need for uniqueness tend to perceive virtual influencers as less distant, ultimately leading to more significant behavioral outcomes (Sands et al., 2022). These consumers are likely among those who strongly recognize virtual influencers' distinctiveness compared to humans and, consequently, may perceive virtual influencers to be unique.

Furthermore, consistent with previous studies, we found that consumers have ambivalent feelings toward virtual influencers (Li et al., 2023; Wan & Jiang, 2023; Sands et al., 2022). This is reflected as the positive (negative) perception of uniqueness (trustworthiness) when comparing the behavioral outcomes for virtual and human influencers. However, instead of separately presenting these perceptions, we integrated them into a single model. This allowed us to confirm that positive perceptions, such as uniqueness, can offset negative ones, such as trustworthiness (Study 2).

Lastly, we showed that while influencers' perceived uniqueness and trustworthiness impact consumers' purchase intention and intention to follow, this effect depends on consumers' ability to identify with the influencer (Study 3). This is an important downstream effect consistently highlighted in the influencer marketing literature (Carlson et al., 2020; Hu et al., 2020; Janssen et al., 2022; Lou & Yuan, 2019; Schouten et al., 2019). Moreover, this relationship holds for both human and virtual influencers. When consumers cannot relate to the influencer, the effect of perceived uniqueness and trustworthiness on purchase intentions weakens, and neither factor significantly affects purchase intentions. However, when consumers strongly identify with the influencer, these perceptions become even more influential, leading to positive behavioral outcomes. Notably, these perceptions can enhance consumers' intention to follow the influencer, irrespective whether consumers show high or low identification with the influencer.

4.1. Theoretical and practical implications

Our research contributes to theory in several ways. First, it expands the literature on consumer responses to (hyper-realistic) virtual influencers. Many studies have concentrated on negative perceptions, such as distrust, limitations in sensory capabilities, credibility, and the perceived increase in social distance. These often drive a preference for human influencers in advertising campaigns (Li et al., 2023; Sands et al., 2022; Stein et al., 2022; Franke et al., 2022; Wan & Jiang, 2023). However, we adopt a more positive perspective, revealing virtual influencers' effectiveness in advertising campaigns. Moreover, we contribute by adding a positive aspect: the perception of uniqueness. We demonstrate that the distinctive characteristics which distinguish virtual influencers from humans, such as novelty, customization, flexibility, and the blending of mixed reality, contribute to the former's individuation or perceived uniqueness. This perception not only draws attention to virtual influencers (de Brito Silva et al., 2022; Franke et al., 2022; Lou et al., 2022), but also leads to other behavioral outcomes, including increased purchase intention towards the endorsed product/brand and a greater intention to follow the influencer.

We further contribute to the literature on consumer responses to (hyper-realistic) virtual influencers by identifying ambivalent perceptions toward virtual influencers. Unlike studies that predominantly considered one-sided or univalent perceptions when studying virtual influencers (Li et al., 2023; Wan & Jiang, 2023; Sands et al., 2022), we explore the coexistence of positive and negative viewpoints towards virtual influencers. Consumers may simultaneously hold favorable (e.g., regarding virtual influencers as unique) and less favorable perceptions (e.g., doubts about their trustworthiness). This suggests that virtual influencers' positive perceptions may outweigh the negative ones, thereby making them more effective than human influencers in specific situations for promoting products, brands, and attracting followers on social media.

Last, we make a valuable contribution to the literature on consumer-endorser identification in influencer marketing. Human influencer marketing research consistently underscores consumer-endorser identification's importance as a pivotal factor influencing behavioral outcomes (Carlson et al., 2020; Hu et al., 2020; Janssen et al., 2022; Lou & Yuan, 2019; Schouten et al., 2019). Likewise, our findings indicate that consumer-endorser identification plays a significant role in driving positive consumer responses to virtual influencers. This heightened identification amplifies consumers' perception of virtual influencers' uniqueness, thereby bolstering positive behavioral outcomes.

Practically, given that virtual influencers can be highly effective due to their perceived uniqueness, we suggest several strategies for marketers and agencies to leverage their potential in campaigns. First, highlighting virtual influencers' distinctiveness from their human counterparts can significantly enhance the former's effectiveness in marketing campaigns. This can be achieved by incorporating these unique traits into the campaign narratives and content. For instance, collaborations with both human and virtual influencers who seamlessly blend elements of the virtual and real worlds while sharing values with the brand can make influencer marketing more compelling.

Moreover, the novelty factor, which contributes to their uniqueness, may diminish over time and potentially make them less exciting in advertisements. However, if advertisements emphasize the ongoing evolution of customization and flexibility in virtual influencers, these aspects can continue to capture attention in marketing materials. This may ensure that the sense of uniqueness remains strong and continues to motivate people to take action, such as making a purchase or following the brand.

Additionally, specific market segments need to be identified where the perception of uniqueness is likely to positively resonate. Tailoring campaigns to these segments and highlighting virtual influencers' unique attributes is key. For instance, younger demographics and subcultures, like anime enthusiasts, often appreciate uniqueness and individuality. Therefore, virtual influencers who align with these audiences' specific interests and passions are likely to be particularly appealing to them.

Lastly, as strong consumer-influencer identification and high perceived uniqueness can lead to even more favorable behavioral outcomes, we recommend practices such as fostering a strong sense of identification between the target audience and virtual influencers. Encouraging consumers to relate to and connect with these digital personalities is essential for building trust and engagement. For instance, this connection can be facilitated by ensuring that the virtual influencer-generated content aligns with the target audience's values and interests. This can make it easier for consumers to relate to the influencer on a personal level.

4.2. Limitations and future research directions

This research emphasizes that the perception of uniqueness plays a crucial role in enhancing virtual influencers' brand endorsement effectiveness compared to their human counterparts. We argue that this perception of uniqueness is based on characteristics which distinguish virtual influencers from humans, including technological advancements that pique curiosity, customization, flexibility, or the seamless blending of physical and virtual realities. These factors have the potential to influence perceptions of uniqueness driving positive consumer behavior toward virtual influencer-endorsed products and brands (Franke et al., 2022; Lou et al., 2022;). However, a pertinent question arises: How long will this positive perception of uniqueness persist? As consumers become accustomed to these unique characteristics, will they continue to drive positive behavioral outcomes? A report from Hyper Auditor suggests that 57 % of virtual influencers' accounts are losing followers (Baklanov, 2024). This may be because these influencers are arousing less curiosity and becoming less of a novelty. Still, other unique attributes can be considered, especially customization and flexibility. By consistently evolve in these aspects, virtual influencers may be able to maintain their distinctiveness compared to humans. Consequently, we strongly encourage researchers to delve deeper into customization and flexibility. Understanding virtual influencers' enduring impact on consumer behavior may underscore their effectiveness compared with human influencers in the long term.

Furthermore, we investigated how the perception of an influencer's uniqueness impacts specific dependent variables, especially purchase intention and the intention to follow, as behavioral outcomes. Scholars may also examine additional dependent variables, such as whether perceptions about the influencer affect how consumers perceive the brand, including aspects like brand innovativeness. For instance, research shows that innovative brands are associated with various factors, including uniqueness (Shams et al., 2015). Then, can the perception of uniqueness attributed to virtual influencers be extended to a brand that aspires to be viewed as innovative, thus influencing how consumers perceive that brand? Hence, we encourage researchers to explore whether an influencer's perception can transfer to a brand or product.

5. Conclusion

We found that virtual influencers can effectively influence consumers' purchase intention and intention to follow the former when consumers perceive virtual influencers to be more unique than human influencers (Studies 1a and 1b). Additionally, we found a parallel mediation effect: virtual (versus human) influencers positively (negatively) impact behavioral outcomes via increased perceived uniqueness (decreased perceived trustworthiness) (Studies 2 and 3). Furthermore, both the perceptions of uniqueness and trustworthiness are strengthened by the degree of consumer identification with the influencer (Study 3). Specifically, perceived uniqueness increased when consumers had a stronger identification with the virtual influencer than the human influencer. The same was observed for perceived trustworthiness.

Declaration of Generative AI and AI-assisted technologies in the writing process

While preparing this work, the authors used ChatGPT to improve the language and readability of sentences, and check grammar and spelling. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication

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CRediT authorship contribution statement

Barbara Kobuszewki Volles: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jaewoo Park:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization. **Anneleen Van Kerckhove:** Visualization, Validation, Supervision, Resources. **Maggie Geuens:** Visualization, Validation, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

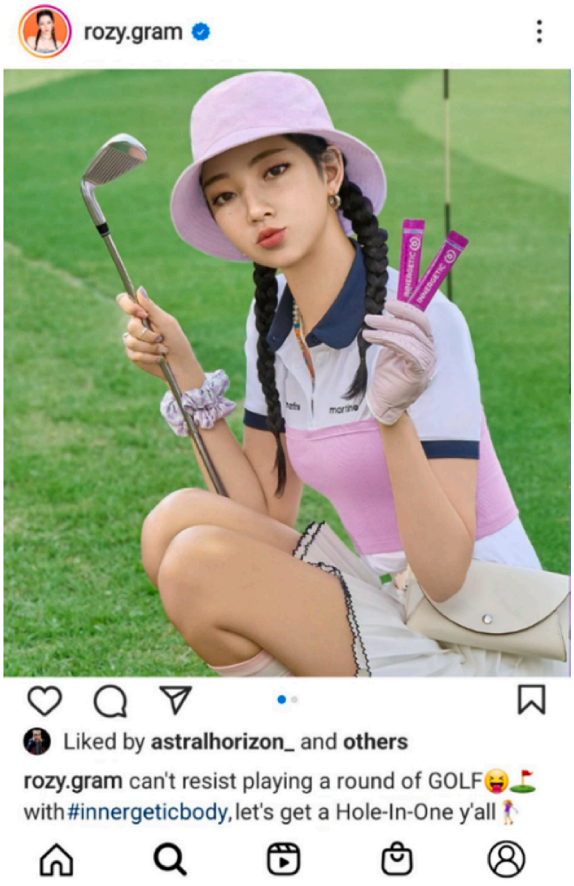
Study 1a – Scenario and Stimuli

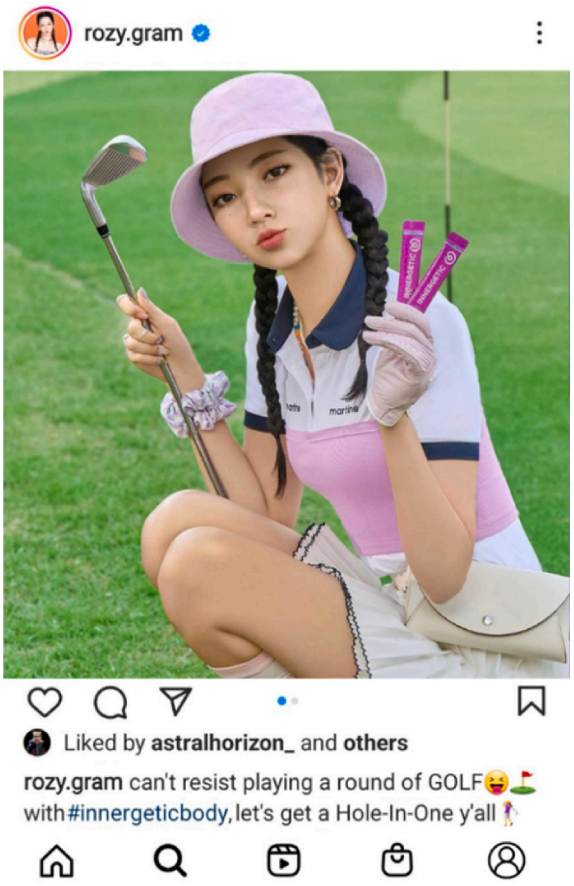
Condition: human influencer	Condition: virtual influencer
There are different types of influencers endorsing brands on social media, from celebrities to influencers that emerged from social media, such as vloggers, TikTokers, and Instafamous personalities. Recently, a new type of influencer has arisen: the virtual influencer. These influencers are computer-generated characters with physical characteristics, features, and personalities of humans. Although they are not humans, their content is produced by artificial intelligence algorithms to mimic human behavior. In the same way, they entertain people on social media as humans do. Both virtual and human influencers are endorsing brands. For example, Rozy is a *human influencer*, one of the Instafamous personalities that emerged from social media. Rozy has been an influencer for three years. She has a few hundred thousand followers on Instagram. Rozy was named one of the Top 50 Emerging Social Media Influencers of 2022. She is known for sharing an inspirational lifestyle and for promoting a range of products and brands. This week, Rozy promoted an energy bar called Innergetic Body	There are different types of influencers endorsing brands on social media, from celebrities to influencers that emerged from social media, such as vloggers, TikTokers, and Instafamous personalities. Recently, a new type of influencer has arisen: the virtual influencer. These influencers are computer-generated characters with physical characteristics, features, and personalities of humans. Although they are not humans, their content is produced by artificial intelligence algorithms to mimic human behavior. In the same way, they entertain people on social media as humans do. Both virtual and human influencers are endorsing brands. For example, Rozy is a *virtual influencer*, one of the computer-generated characters. Rozy has been an influencer for three years now. She has a few hundred thousand followers on Instagram. Rozy was named one of the Top 50 Emerging Social Media Influencers of 2022. She is known for sharing an inspirational digital lifestyle and for promoting a range of products and brands. This week, Rozy promoted an energy bar called Innergetic Body on her social media account. Please have a good look at her post below before answering the questions.

(*continued on next page*)

(*continued*)

Condition: human influencer	Condition: virtual influencer
on her social media account. Please have a good look at her post below before answering the questions.	





Study 1b – Scenario and Stimuli

Condition: human influencer	Condition: virtual influencer
*Emma Carlson is a social media influencer.*Emma Carlson has been an influencer for five years and has a few hundred thousand followers on Instagram. As an influencer, @emmacarlson promotes a range of products and brands, such as Porsche, Gucci, Louis Vuitton, and Dior, and shares brand and product experiences with her followers. Emma Carlson is known for promoting luxury brands and sharing inspirational and creative posts. She was named one of the Top 50 Emerging Social Media Influencers of 2022. This week, Emma Carlson posted a picture endorsing a bag from Louis Vuitton. Please have a careful look at the picture the influencer posted. We will next ask you questions about it.	*Emma Carlson is a social media influencer powered by artificial intelligence (AI). This means the influencer only exists digitally on social media and is designed using computer 3D graphics to look and behave like a human.*Emma Carlson has been an influencer for five years and has a few hundred thousand followers on Instagram. As an influencer, @emmacarlson promotes a range of products and brands, such as Porsche, Gucci, Louis Vuitton, and Dior, and shares brand and product experiences with her followers. Emma Carlson is known for promoting luxury brands and sharing inspirational and creative posts. She was named one of the Top 50 Emerging Social Media Influencers of 2022. This week, Emma Carlson posted a picture endorsing a bag from Louis Vuitton. Please have a careful look at the picture the influencer posted. We will next ask you questions about it.

(*continued on next page*)

(*continued*)

Condition: human influencer	Condition: virtual influencer
	

Study 2 – Scenario and Stimuli

Condition: human influencers	Condition: virtual influencers
Brad Simmons is a social media influencer. Brad Simmons has been an influencer for five years and has a few hundred thousand followers on Instagram. As an influencer, Brad Simmons promotes a range of brands, such as Porsche, Louis Vuitton, and Armani, and shares brand experiences with his followers. Brad Simmons is known for promoting luxury brands and sharing an inspirational lifestyle. He was named one of the Top 50 Emerging Social Media Influencers of 2022. This week, Brad Simmons posted a picture endorsing a shirt from Armani. Please have a careful look at the picture the influencer posted. Next, we will ask you questions about it.	*Brad Simmons is a virtual influencer on social media powered by artificial intelligence (AI). This means the influencer only exists digitally on social media and is designed using computer 3D graphics to look and behave like a human.* Brad Simmons has been an influencer for five years and has a few hundred thousand followers on Instagram. As an influencer, Brad Simmons promotes a range of brands, such as Porsche, Louis Vuitton, and Armani, and shares brand experiences with his followers. Brad Simmons is known for promoting luxury brands and sharing an inspirational digital lifestyle. He was named one of the Top 50 Emerging Social Media Influencers of 2022. This week, Brad Simmons posted a picture endorsing a shirt from Armani. Please have a careful look at the picture the influencer posted. Next, we will ask you questions about it.

(*continued on next page*)

(*continued*)

Condition: human influencers	Condition: virtual influencers
	

Study 3 – Scenario and Stimuli

Condition: human influencer	Condition: virtual influencer
There are different types of influencers endorsing brands on social media, from celebrities to influencers that emerged from social media, such as vloggers, TikTokers, and Instafamous personalities. Besides these well-known types of human influencers, a new type of influencer has emerged: the virtual influencer. These influencers are non-human, digitally created characters sharing social media content and engaging in interactive communications to obtain an influential status among consumers. They have physical characteristics, features, and personalities similar to humans. Both human and virtual influencers are endorsing brands. For example, Daisy is a *human influencer* and one of the Instafamous personalities. Daisy has been an influencer for three years now, with a few hundred thousand followers on Instagram. Daisy was named one of the Top 50 Emerging Social Media Influencers of 2022.She is known for sharing an inspirational digital lifestyle and for promoting a range of products and brands. Below, you can see an example of a post from Daisy of herself at an event promoting a bag from Gucci on her social media account.	There are different types of influencers endorsing brands on social media, from celebrities to influencers that emerged from social media, such as vloggers, TikTokers, and Instafamous personalities. Besides these well-known types of human influencers, a new type of influencer has emerged: the virtual influencer. These influencers are non-human, digitally created characters sharing social media content and engaging in interactive communications to obtain an influential status among consumers. They have physical characteristics, features, and personalities similar to humans. Both human and virtual influencers are endorsing brands. For example, Daisy is a *virtual influencer* and one of the digitally created characters. Daisy has been an influencer for three years now, with a few hundred thousand followers on Instagram. Daisy was named one of the Top 50 Emerging Social Media Influencers of 2022.She is known for sharing an inspirational digital lifestyle and for promoting a range of products and brands. Below, you can see an example of a post from Daisy of herself at an event promoting a bag from Gucci on her social media account.

(*continued on next page*)

(*continued*)

Condition: human influencer	Condition: virtual influencer
	

Appendix B

Independent *t*-test to analyze the direct impact of the independent variable (influencer type: human vs. virtual) on the dependent variables (purchase intention and intention to follow) in the absence of any mediator, across studies 1a-1b, study 2, and study 3.

	Mediator and Dependent Variables	Condition Human Influencer	Condition Virtual Influencer	Statistical test
Study 1a	Variables	Mean (SD)	Mean (SD)	t(df), Sig. (2-tailed)
		$N = 57$	$N = 63$	
	Purchase Intention	2.96 (1.76)	2.70 (1.61)	$t(118) = 0.865$, $p = 0.389$
	Intention to follow	2.19 (1.56)	2.10 (1.65)	$t(118) = 0.332$, $p = 0.741$
	Perceived Uniqueness	2.70 (1.20)	3.50 (1.54)	$t(118) = -3.13$, $p = 0.002$
Study 1b	Variables	Mean (SD)	Mean (SD)	t(df), Sig. (2-tailed)
		$N = 106$	$N = 113$	
	Purchase Intention	2.38 (1.72)	2.46 (1.59)	$t(217) = -0.369$, $p = 0.712$
	Intention to follow	1.97 (1.56)	2.04 (1.48)	$t(217) = -0.352$, $p = 0.725$
	Perceived Uniqueness	2.38 (1.36)	3.38 (1.57)	$t(217) = -5.04$, $p < 0.001$
Study 2	Variables	Mean (SD)	Mean (SD)	t(df), Sig. (2-tailed)
		$N = 160$	$N = 182$	
	Purchase Intention	2.18 (1.49)	2.32 (1.66)	$t(173) = -0.585$, $p = 0.560$
	Intention to follow	2.01 (1.42)	2.16 (1.67)	$t(173) = -0.636$, $p = 0.526$
	Perceived Uniqueness	2.51 (1.39)	3.77 (1.56)	$t(173) = -5.62$, $p < 0.001$
	Perceived Trustworthiness	2.51 (1.39)	3.77 (1.56)	$t(173) = 1.66$, $p = 0.098$
Study 3	Variables	Mean (SD)	Mean (SD)	t(df), Sig. (2-tailed)
		$N = 160$	$N = 182$	
	Purchase Intention	2.03 (1.51)	2.09 (1.59)	$t(340) = -0.40$, $p = 0.686$
	Intention to follow	2.36 (1.60)	2.10 (1.63)	$t(340) = 1.50$, $p = 0.134$
	Perceived Uniqueness	2.53 (1.35)	3.57 (1.66)	$t(340) = -6.23$, $p < 0.001$
	Perceived Trustworthiness	3.91(1.26)	3.27(1.64)	$t(340) = 6.04$, $p < 0.001$

References

Alain, T. (2024, July 15). Influencer engagement: Everything you need to know. Upfluence. https://www.upfluence.com/influencer-marketing/influencer-engagement-everything-you-need-to-know#:~:text = Most%20influencers%20will%20have%20an,above%20shows%20a%20strong%20performance.

Arsenyan, J., & Mirowska, A. (2021). Almost human? A comparative case study on the social media presence of virtual influencers. *International Journal of Human-Computer Studies, 155*, Article 102694. https://doi.org/10.1016/J.IJHCS.2021.102694

Baklanov, N. (2024, July 15). The top virtual Instagram influencers in 2022. Hype Auditor. https://hypeauditor.com/blog/the-top-virtual-instagram-influencers-in-2022/.

Block, E., & Lovegrove, R. (2021). Discordant storytelling, 'honest fakery', identity peddling: How uncanny CGI characters are jamming public relations and influencer practices. *Public Relation Inquiry, 10*(3), 265–293. https://doi.org/10.1177/2046147X211026936

Byun, K. J., & Ahn, S. J. (2023). A systematic review of virtual influencers: Similarities and differences between human and virtual influencers in interactive advertising. *Journal of Interactive Advertising, 23*(4), 1–14. https://doi.org/10.1080/15252019.2023.2236102

Carlson, B. D., Donavan, D. T., Deitz, G. D., Bauer, B. C., & Lala, V. (2020). A customer-focused approach to improve celebrity endorser effectiveness. *Journal of Business Research, 109*, 221–235. https://doi.org/10.1016/j.jbusres.2019.11.048

Casaló, L. V., Flavián, C., & Ibáñez-Sánchez, S. (2020). Influencers on Instagram: Antecedents and consequences of opinion leadership. *Journal of Business Research, 117*, 510–519. https://doi.org/10.1016/j.jbusres.2018.07.005

Choi, S. M., & Rifon, N. J. (2012). It is a match: The impact of congruence between celebrity image and consumer ideal self on endorsement effectiveness. *Psychology & Marketing, 29*(9), 639–650. https://doi.org/10.1002/mar.20550

Chung, S., & Cho, H. (2017). Fostering parasocial relationships with celebrities on social media: Implications for celebrity endorsement. *Psychology & Marketing, 34*(4), 481–495. https://doi.org/10.1002/mar.21001

Conti, M., Gathani, J., & Tricomi, P. P. (2022). Virtual influencers in online social media. *IEEE Communications Magazine, 60*(8), 86–91. https://doi.org/10.1109/MCOM.001.2100786

de Brito Silva, M. J., de Oliveira Ramos Delfino, L., Alves Cerqueira, K., & de Oliveira Campos, P. (2022). Avatar marketing: A study on the engagement and authenticity of virtual influencers on Instagram. *Social Network Analysis and Mining, 12*(1), 130. https://doi.org/10.1007/s13278-022-00966-w

Fakhreddin, F., & Foroudi, P. (2022). Instagram influencers: The role of opinion leadership in consumers' purchase behavior. *Journal of Promotion Management, 28*(6), 795–825. https://doi.org/10.1080/10496491.2021.2015515

Farivar, S., Wang, F., & Yuan, Y. (2023). Influencer marketing: A perspective of the elaboration likelihood model of persuasion. *Journal of Electronic Commerce Research, 24*(2), 127–145.

Farivar, S., & Wang, F. (2021). Influencer marketing: Current knowledge and research agenda. In *Digital Marketing & eCommerce Conference* (pp. 201–208). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-76520-0_21.

Franke, C., Groeppel-Klein, A., & Müller, K. (2022). Consumers' responses to virtual influencers as advertising endorsers: Novel and effective or uncanny and deceiving? *Journal of Advertising, 52*(4), 1–17. https://doi.org/10.1080/00913367.2022.2154721

Gambino, A., Fox, J., & Ratan, R. A. (2020). Building a stronger CASA: Extending the computers are social actors paradigm. *Human-Machine Communication, 1*, 71–85. https://doi.org/10.30658/hmc.1.5

Geyser, W. (2024, July 15). The state of AI in influencer marketing: A comprehensive benchmark report. Influencer Marketing Hub. https://influencermarketinghub.com/ai-in-influencer-marketing/.

Ham, J., Li, S., Shah, P., & Eastin, M. S. (2023). The "mixed" reality of virtual brand endorsers: Understanding the effect of brand engagement and social cues on technological perceptions and advertising effectiveness. *Journal of Interactive Advertising, 23*(2), 98–113. https://doi.org/10.1080/15252019.2023.2185557

Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford Publications.

Hofeditz, L., Nissen, A., Schütte, R., & Mirbabaie, M. (2022). Trust me, I'm an influencer! – A comparison of perceived trust in human and virtual influencers. ECIS 2022 Research-in-Progress Papers. https://aisel.aisnet.org/ecis2022_rip/27.

Hoffner, C., & Buchanan, M. (2005). Young adults' wishful identification with television characters: The role of perceived similarity and character attributes. *Media Psychology, 7*(4), 325–351. https://doi.org/10.1207/S1532785XMEP0704_2

Hoffner, C. (1996). Children's wishful identification and parasocial interaction with favorite television characters. *Journal of Broadcasting & Electronic Media, 40*(3), 389–402. https://doi.org/10.1080/08838159609364360

Horton, D., & Wohl, R. (1956). Mass communication and para-social interaction: Observations on intimacy at a distance. *Psychiatry, 19*(3), 215–229. https://doi.org/10.1080/00332747.1956.11023049

Hu, L., Min, Q., Han, S., & Liu, Z. (2020). Understanding followers' stickiness to digital influencers: The effect of psychological responses. *International Journal of Information Management, 54*, Article 102169. https://doi.org/10.1016/j.ijinfomgt.2020.102169

Janssen, L., Schouten, A. P., & Croes, E. A. (2022). Influencer advertising on Instagram: Product-influencer fit and number of followers affect advertising outcomes and influencer evaluations via credibility and identification. *International Journal of Advertising, 41*(1), 101–127. https://doi.org/10.1080/02650487.2021.1994205

Kim, S. Y., Schmitt, B. H., & Thalmann, N. M. (2019). Eliza in the uncanny valley: Anthropomorphizing consumer robots increases their perceived warmth but decreases liking. *Marketing Letters, 30*, 1–12. https://doi.org/10.1007/s11002-019-09485-9

Lee, J. A., & Eastin, M. S. (2021). Perceived authenticity of social media influencers: Scale development and validation. *Journal of Research in Interactive Marketing, 15*(4), 822–841. https://doi.org/10.1108/JRIM-12-2020-0253

Li, H., Lei, Y., Zhou, Q., & Yuan, H. (2023). Can you sense without being human? Comparing virtual and human influencers endorsement effectiveness. *Journal of Retailing and Consumer Services, 75*, Article 103456. https://doi.org/10.1016/j.jretconser.2023.103456

Lou, C., & Yuan, S. (2019). Influencer marketing: How message value and credibility affect consumer trust of branded content on social media. *Journal of Interactive Advertising, 19*(1), 58–73. https://doi.org/10.1080/15252019.2018.1533501

Lou, C., Kiew, S. T. J., Chen, T., Lee, T. Y. M., Ong, J. E. C., & Phua, Z. (2022). Authentically fake? How consumers respond to the influence of virtual influencers. *Journal of Advertising, 52*(4), 540–557. https://doi.org/10.1080/00913367.2022.2149641

Maslach, C., Stapp, J., & Santee, R. T. (1985). Individuation: Conceptual analysis and assessment. *Journal of Personality and Social Psychology, 49*(3), 729–738. https://doi.org/10.1037/0022-3514.49.3.729

Mori, M. (1970). The uncanny valley: The original essay by Masahiro Mori. *IEEE Spectrum, 6*.

Mouritzen, S. L. T., Penttinen, V., & Pedersen, S. (2023). Virtual influencer marketing: The good, the bad and the unreal. *European Journal of Marketing*. https://doi.org/10.1108/EJM-12-2022-0915

Ohanian, R. (1990). Construction and validation of a scale to measure celebrity endorsers' perceived expertise, trustworthiness, and attractiveness. *Journal of Advertising, 19*(3), 39–52. https://doi.org/10.1080/00913367.1990.10673191

Reeves, B., & Nass, C. (1996). The media equation: How people treat computers, television, and new media like real people and places. Cambridge.

Sands, S., Campbell, C. L., Plangger, K., & Ferraro, C. (2022). Unreal influence: Leveraging AI in influencer marketing. *European Journal of Marketing, 56*(6), 1721–1747. https://doi.org/10.1108/EJM-12-2019-0949

Schouten, A. P., Janssen, L., & Verspaget, M. (2019). Celebrity vs. influencer endorsements in advertising: The role of identification, credibility, and product-endorser fit. *International Journal of Advertising, 39*(2), 258–281. https://doi.org/10.1080/02650487.2019.1634898

Shams, R., Alpert, F., & Brown, M. (2015). Consumer perceived brand innovativeness: Conceptualization and operationalization. *European Journal of Marketing, 49*(9/10), 1589–1615. https://doi.org/10.1108/EJM-05-2013-0240

Stein, J. P., Breves, P. L., & Anders, N. (2022). Parasocial interactions with real and virtual influencers: The role of perceived similarity and human-likeness. *New Media & Society*. https://doi.org/10.1177/14614448221102900

Tajfel, H., & Turner, J. C. (2004). The social identity theory of intergroup behavior. In *Political psychology* (pp. 276–293). Psychology Press.

Tinwell, A., Nabi, D. A., & Charlton, J. P. (2013). Perception of psychopathy and the uncanny valley in virtual characters. *Computers in Human Behavior, 29*(4), 1617–1625. https://doi.org/10.1016/j.chb.2013.01.008

Vigneron, F., & Johnson, L. (2004). Measuring perceptions of brand luxury. *Journal of Brand Management, 11*, 484–506. https://doi.org/10.1057/palgrave.bm.2540194

Wan, A., & Jiang, M. (2023). Can virtual influencers replace human influencers in live-streaming e-commerce? An exploratory study from practitioners' and consumers' perspectives. *Journal of Current Issues & Research in Advertising, 44*(3), 332–372. https://doi.org/10.1080/10641734.2023.2224416

Xie-Carson, L., Magor, T., Benckendorff, P., & Hughes, K. (2023). All hype or the real deal? Investigating user engagement with virtual influencers in tourism. *Tourism Management, 99*, Article 104779. https://doi.org/10.1016/j.tourman.2023.104779

Xu, X., & Pratt, S. (2018). Social media influencers as endorsers to promote travel destinations: An application of self-congruence theory to the Chinese Generation Y. *Journal of Travel & Tourism Marketing, 35*(7), 958–972. https://doi.org/10.1080/10548408.2018.1468851

Yang, J., Chuenterawong, P., Lee, H., & Chock, T. M. (2023). Anthropomorphism in CSR endorsement: A comparative study on humanlike vs. cartoonlike virtual influencers' climate change messaging. *Journal of Promotion Management, 29*(5), 705–734. https://doi.org/10.1080/10496491.2022.2163041

Barbara Kobuszewski Volles holds a PhD from the Faculty of Economics and Business Administration, Department of Marketing, Innovation, and Organization at Ghent University. Her research interests lie in digital marketing, with a specific focus on consumers' judgments and decisions regarding marketing tactics for promoting products and brands.

Jaewoo Park is a Professor of Marketing at the Faculty of Commerce, Department of Marketing and Internationl Trade of Chuo University (Japan). His research interests include sensory marketing, design aesthetics, luxury perception, and branding. His scholarly work has been published in academic journals such as Journal of Consumer Psychology, Journal of Retailing, Journal of Advertising, Psychology & Marketing, Journal of Product and Brand Management, Food Quality and Preference, and Journal of Business Research.

Anneleen Van Kerckhove is Associate Professor of Marketing at the Faculty of Economics and Business Administration, Department of Marketing, Innovation, and Organization at Ghent University. She previously obtained her PhD also from Ghent University. Two key research interests pertain to the influences of external situations on consumers' judgments and decisions and of marketing elements on health behavior. Both domains have clear societal relevance, leading to the publication of her research in, among others, Journal of

Consumer Research, Journal of Marketing Research, Journal of Consumer psychology, International Journal of Research in Marketing, and Journal of Business Research.

Maggie Geuens is Professor of Marketing at the Faculty of Business Economics, Department of Marketing, Innovation, and Organization of Ghent University. Her main research interests involve marketing communications and consumer behavior, with a special focus on healthy and sustainable food consumption decisions. Her work has been published in a variety of international journals, including Journal of Consumer Research, Journal of Consumer Psychology, Journal of Marketing Research, International Journal of Research in Marketing, Journal of Advertising, Journal of Business Research, Food Quality and Preference, and Journal of Public Policy and Marketing. She serves on the editorial boards of several journals and is co-author of a textbook on Marketing Communications (with Patrick De Pelsmacker and Joeri Van den Bergh).